HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Assisted with the planning and design of port infrastructure, including quay wall structures and waterfront developments, for major European port authorities.
- Produced technical reports, specifications, and feasibility studies for coastal resilience projects, addressing sea level rise and flood relief schemes.
- Collaborated with a multidisciplinary team of engineers and stakeholders to develop practical guidance on coastal erosion mitigation strategies.
- Undertook site visits and surveys to gather data for maritime infrastructure projects, meeting with clients and local authorities to understand project requirements.
- Developed Python scripts to automate data extraction and analysis for environmental impact assessments related to coastal developments.
- Managed change control processes for system configuration updates and coordinated User Acceptance Testing (UAT) for integrated enterprise systems.
- Containerised Java Spring Boot applications using Docker and deployed them to AWS EC2 instances, managing application lifecycles and basic scaling.

SKILLS

Engineering Principles – Maritime Engineering, Civil Engineering, Coastal Engineering, Structural Engineering, Geotechnical Engineering

Software Development – Java, Spring Boot, Python, REST APIs, Docker, Git

Data Analysis & BI – SQL, PostgreSQL, MySQL, Oracle, Power BI, Tableau, DAX, Power Query

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Project Management & Tools – Jira, Confluence, Jenkins, Agile, Scrum

PROJECTS

Coastal Resilience Study: Sea Level Rise Impact

- Developed Python scripts to process and analyze historical tidal data and meteorological records for a coastal resilience study.
- Created data visualizations and summary reports in Power BI to illustrate projected impacts of sea level rise on local infrastructure.
- Collaborated with MSc Data Analytics peers to interpret statistical findings and present them to a simulated client group.

Port Infrastructure Design Simulation

- Modeled structural loads and constraints for a simulated quay wall design using engineering principles derived from coursework.
- Generated technical drawings and specification outlines based on simulated project requirements for a major port expansion.
- Assisted in preparing a business case document outlining potential infrastructure upgrades and their associated benefits.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)